

Experimental Research (C2)

02

Methods and Studies with Users

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Some Questions

- 1. Two types of HCI research most frequently performed?**
- 2. One benefit and one drawback of controlled laboratory studies versus field studies.**

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Last Class Summary

Presentation

Introduction to HCI Research

- Brief history of HCI
- Types of HCI research contributions
- HCI research topics over time
- HCI research methods over time
- How HCI research differs
- The research lifecycle
- Trade-offs in HCI

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Overview

Experimental Research

- Types of behavioral research
- Research hypotheses
- Basics of experimental research
- Significance tests
- Limitations of experimental research

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INTRODUCTION & TYPES OF BEHAVIORAL RESEARCH

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Introduction

**Several laboratory and nonlaboratory methods for
HCIR**

Observations

Field studies

Surveys

Usability studies

Interviews

Focus groups

Controlled experiments

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Introduction

Example: “Study how users enter information into their mobile phones”

Observation (in a natural setting)

Individuals using a cell phone in an airport

Survey

Questions about entering information

Interview

To find out how they enter information

Usability test

Recruit participants to run a usability test in the lab

Etc.

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Introduction

Observation

- + Identification of patterns used in natural settings
- Extremely time consuming

Survey

- + Large number of participants in a short period of time
- Misunderstanding of questions; sample biased

Interviews

- + Clarification of questions and follow-up questions
- Takes more time than surveys

Usability test

- + Quick and low-cost method for identifying key usability problems
- No guarantee that all critical problems can be identified

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Introduction

Method to use is highly context-dependent

Depends on a variety of factors

primary purpose of the study

time constraints

funding

participant pool

researchers' experience

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Introduction

How to select the best research method?

Will be discussed in more detail in the next classes

Today

Experimental research in general

Basics of conducting experimental studies

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Empirical Research – What & Why?

Empirical research is

observation and **measurement-based** investigation
seeking to discover and interpret facts, theories, or
laws (relating to humans interacting with computers)

We conduct empirical research to

Answer (and raise!) questions about new or existing user
interface designs or interaction techniques
Develop or explore models that *describe* or *predict*
behavior (of humans interacting with computers)

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Types of Behavioral Research

Behavioral Research

Combines quantitative and qualitative methods to
measure human behavior

Types of Behavioral Research

Descriptive investigations

Relational investigations

Experimental investigations

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Types of Behavioral Research

Descriptive investigations

Construct an accurate description of what is happening

Example: “Students who play a game write better than those who don’t play”

Provides only an interesting observation

Doesn’t allow to identify any causal relationship

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Types of Behavioral Research

Relational investigations

Identify relations between multiple factors

Can rarely determine the causal relationship between multiple factors

Example: Nr. of hours played vs writing speed

We observe that those who play more write faster

BUT that doesn’t mean that they write faster because they play more

It could also be because (for instance)

Those who write faster like to play the game more

Those who read better tend to write faster, and those who read well like to play the game more

Now what?

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Types of Behavioral Research

Experimental investigations

Identify causal relationship between multiple factors

Example of how to do it:

Two groups of students with the same characteristics
Measure writing speed at the beginning
One group plays the game the other doesn't
After 3 months (for e.g.) we measure the writing speed
If students who play write faster, we can conclude that
playing improves writing speed

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Types of Behavioral Research

Type of Research	Focus	General Claims	Typical Methods
Descriptive	Describe a situation or a set of events	X is happening	Observations, field studies, focus groups, interviews
Relational	Identify relations between multiple variables	X is related to Y	Observations, field studies, surveys
Experimental	Identify causes of a situation or a set of events	X is responsible for Y	Controlled experiments

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02

RESEARCH HYPOTHESES

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Research Hypotheses

An experiment starts with a **research hypothesis**

Hypothesis - precise problem statement

that can be tested through an empirical investigation

A concrete research hypothesis lays

the foundation of an experiment

the basis of statistical significance testing

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Types of Hypotheses

Null Hypothesis (H_0)

typically states that **there is no difference** between experimental treatments / conditions

Alternative Hypothesis (H_1)

a statement that is mutually exclusive with the null hypothesis

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Types of Hypotheses

The goal of an experiment is

- to find statistical evidence
- to refute or nullify the null hypothesis
- in order to support the alternative hypothesis

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Types of Hypotheses - Example

“Pull-down menu or a pop-up menu in the home page of the website?”

H₀: There is no difference between the pull-down menu and the pop-up menu in the time spent locating pages.

H₁: There is a difference between the pull-down menu and the pop-up menu in the time spent locating pages.

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Dependent & Independent Variables

A hypothesis should specify

independent variables

dependent variables

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Dependent & Independent Variables

Independent variables (IV)

Factors of interest or

the possible “cause” of the change in the dependent variable

Are independent of a participant’s behavior

Are the treatments or **conditions** that the researchers can control

Dependent variables (DV)

Outcome or effect of interest

Are dependent on a participant’s behavior or the changes in the IVs

Are usually the **outcomes** that the researchers need to measure

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Dependent & Independent Variables - Example

H₀: There is no difference between the pull-down menu and the pop-up menu in the time spent locating pages.

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Dependent & Independent Variables - Example

H_0 : There is no difference between the
pull-down menu and the pop-up menu
in the time spent locating pages.

Dependent

Independent

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Typical **Independent** variables in HCI

Technology

Types of technology or device (e.g. mouse vs joystick)

Types of design (e.g. contrast, background color)

Users:

age, gender, computer experience, professional domain,
education, culture, motivation, mood, and disabilities

Context of use:

Physical factors (e.g. noise, lighting, temperature)

User status (e.g. seated, walking)

Social factors (e.g. number of people surrounding the user)

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Typical Dependent variables in HCI

Efficiency

task completion time; speed (e.g. wpm); ...

Accuracy / Effectiveness

error rate; ...

Subjective satisfaction

Likert scale ratings; ...

Ease of learning and retention rate

Tasks completed without help; functions remembered after some time

Physical or cognitive demand

NASA task load index, ...

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03

BASICS OF EXPERIMENTAL RESEARCH

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Components of an Experiment

Treatments or conditions

Different techniques, devices, or procedures to compare

Units

Objects to which the experiment treatments are applied

In HCI are normally human subjects

with specific characteristics (e.g. gender, age, or computing experience)

Assignment method

the way in which the different treatments are assigned to the experimental units

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Components of an Experiment - Example

Experiment: Compare typing speed using a traditional QWERTY keyboard and a DVORAK keyboard

Treatments or conditions

Type of keyboard: QWERTY or DVORAK

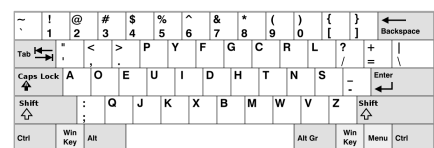
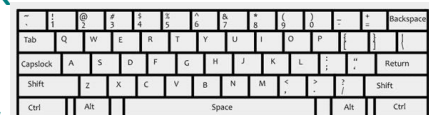
Units

Participants recruited to join the study

Should have no previous experience using either keyboard

Assignment method

Random



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Randomization

Randomization - the random assignment of treatments to the experimental units or participants

In a totally randomized experiment, no one, including the investigators themselves, is able to predict the condition to which a participant is going to be assigned

Methods of randomization

Preliminary methods (e.g. tossing a coin)

Random table

Software driven randomization

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Counter-balancing

Latin Square Design

Counter-balancing conditions

Counter-balancing tasks

	Order of administration			
	1	2	3	4
Sequence 1	A	B	C	D
Sequence 2	B	C	D	A
Sequence 3	C	D	A	B
Sequence 4	D	A	B	C

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SIGNIFICANCE TESTS

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Significance Tests

Almost all experimental investigations are analyzed and reported through significance tests

Example:

*“A t test showed that there was a **significant difference** in the number of lines of text entered ($t(11)=6.28$, $p<0.001$) with more entered in the tactile condition.”*

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Why do we need significance tests?

Can't we just compare means?

E.g. number of errors; or time to complete task

The solution with the smaller values is the better one (??)

Example 1:

Mike's height is 1.88m. Mary's height is 1.73m

So, Mike is taller than Mary. (??) -> true

Values are known, so we can compare and draw conclusions

No uncertainty involved -> no significance test needed

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Why do we need significance tests?

Example 2:

Average height of three males is 1.65m | Average height of three females is 1.78m

So, females are taller than males. (??)

What is wrong with this statement?

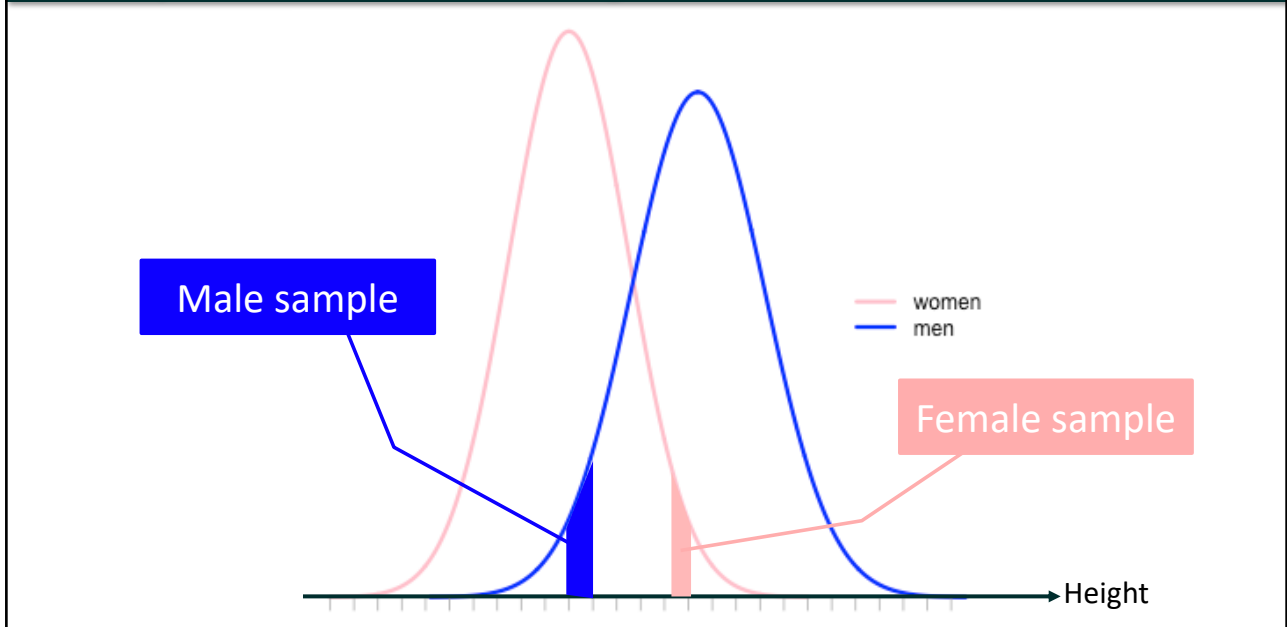
Can find three other males/females with different averages

Groups are too small

Individuals in the groups are not representative

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Why do we need significance tests?



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Why do we need significance tests?

When the **values** of the members of the comparison groups **are all known**, you can directly compare them and draw a conclusion

No significance test is needed since there is no uncertainty involved

When the population is large, we can only **sample a subgroup of people** from the entire population

Significance tests allow us to determine how confident we are that the results observed from the sampling population can be generalized to the entire population

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Type I and Type II errors

Significance testing

process in which a null hypothesis (H_0) is contrasted with an alternative hypothesis (H_1) to determine the **likelihood that the null hypothesis is true**

All significance tests are subject to the risk of Type I and Type II errors

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Type I and Type II errors

Type I error

(also called an α error or a “**false positive**”)
refers to the mistake of rejecting the null hypothesis when it is true and should not be rejected

Type II error

(also called a β error or a “**false negative**”)
refers to the mistake of not rejecting the null hypothesis when it is false and should be rejected

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Type I and Type II errors - Example

Judicial Case

H_0 : Defendant innocent

H_1 : Defendant guilty

Reality	Jury Decision	
	Innocent	Guilty
Innocent	✓	Type I error
Guilty	Type II error	✓

Type I error = False Positive

Type II error = False Negative

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Type I and Type II errors - Example

HCI Experiment (ATMs with buttons vs ATMs with touch screen)

H_0 : No difference in easy of use between b-ATMs and ts-ATMs

H_1 : Touch screen ATMs are easier to use than ATMs with buttons

Reality	Study Conclusion	
	No Difference	Ts-ATMs Easier to Use
No Difference	✓	Type I error
Ts-ATMs Easier to Use	Type II error	✓

Type I error = False Positive

Type II error = False Negative

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Type I and Type II errors

It is generally believed that **Type I errors are worse** than Type II errors

Statisticians call Type I errors a mistake that involves “gullibility”

A Type I error may result in a condition worse than the current state

Type II errors are mistakes that involve “blindness”

A Type II error can cost the opportunity to improve the current state

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Controlling risks of errors

When designing experiments and analyzing data, you have to evaluate the risk of making Type I and Type II errors.

Probability of making a Type I error is called **alpha**
(or significance level, p value)

Probability of making a Type II error is called **beta**

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Controlling risks of errors

Alpha and beta are interrelated

Under the same conditions, decreasing alpha reduces the chance of making Type I errors but increases the chance of making Type II errors

To control the occurrence of Type I errors

Use a very low p value (0.05)

To reduce Type II errors

Use a relatively large sample size

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Limitations of Experimental Research

Difficulty in identifying a testable hypothesis

Limited number of dependent and independent variables

Difficulty in controlling potential confounding factors

They may influence the dependent variables

Their impact can be reduced through well-designed and implemented experiments

Changes in lab-observed behavior as compared to behavior in a more realistic setting

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Summary

Experimental Research

Types of behavioral research

Research hypotheses

Basics of experimental research

Significance tests

Limitations of experimental research

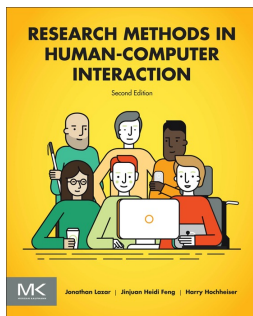
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Additional Information

- **Empirical Research: Definition, Methods, Types and Examples**
 - <https://www.questionpro.com/blog/empirical-research/>
- **Types of Research Designs Compared | Guide & Examples**
 - <https://www.scribbr.com/methodology/types-of-research/>
- **Guide to Experimental Design**
 - <https://www.scribbr.com/methodology/experimental-design/>
- **Hypothesis**
 - <https://www.scribbr.com/methodology/hypothesis/>

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Credits



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